- Let $\vec{a}$ be a vector and m be a scalar. Then the vector $m\vec{a}$ is called the **scalar multiple of a vector** $\vec{a}$ by the scalar m.
- Two vectors $\vec{a}$ and $\vec{b}$ are said to be **parallel** if $\vec{a} = \lambda\vec{b}$, where λ is a scalar.
- If $\vec{a}, \vec{b}$ and $\vec{c}$ are the sides of a triangle taken in order then $\vec{a} + \vec{b} + \vec{c} = \vec{0}$
- Vector addition is associative.
- For any vector $\vec{a}$, $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$.
- For any vector $\vec{a}$, $\vec{a} + (-\vec{a}) = (-\vec{a}) + \vec{a} = \vec{0}$.
- Vector addition is commutative.
- "If two vectors are represented in magnitude and direction by the two sides of a triangle taken in the same order, then their sum is represented by the third side taken in the reverse order". This is known as the **triangle law of addition.**
- In a parallelogram $OABC$, if $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represents two adjacent sides, then the diagonal $\overrightarrow{OC}$ represents their sum. This is **parallelogram law of addition**.
- If α, β, γ are the direction angles then $\cos\alpha, \cos\beta, \cos\gamma$ are the direction cosines.
- The direction ratios of the vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ are x, y, z.
- If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors in the space, then any vector in the space can be written as $l\vec{a} + m\vec{b} + n\vec{c}$ in a unique way.

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of any point and let α, β, γ be the direction angles of $\vec{r}$. Then

(i) the sum of the squares of the direction cosines of $\vec{r}$ is 1.

(ii) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

(iii) the direction cosines of $\vec{r}$ are $\frac{x}{\sqrt{x^2+y^2+z^2}}, \frac{y}{\sqrt{x^2+y^2+z^2}}, \frac{z}{\sqrt{x^2+y^2+z^2}}$.

(iv) l, m, n are the direction cosines of a vector if and only if $l^2 + m^2 + n^2 = 1$.

(v) any unit vector can be written as $\cos\alpha\hat{i} + \cos\beta\hat{j} + \cos\gamma\hat{k}$.

- The **scalar product** of the vectors $\vec{a}$ and $\vec{b}$ is $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$.
- **Vector product** of any two non-zero vectors $\vec{a}$ and $\vec{b}$ is written as $\vec{a} \times \vec{b}$ and is defined as $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}$, where θ is the angle between $\vec{a}$ and $\vec{b}, 0 \le \theta \le \pi$. Here $\vec{a}, \vec{b}, \hat{n}$ form a right handed system.

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